

gives $q = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

17.

Ans: B

Adjacent antinodes are half a wavelength apart. Hence a node is a quarter of a wavelength from an adjacent antinode.

A: W and Z are nodes, hence the kinetic energies of the particles at those positions are zero.

C: X and Y are in adjacent segments hence they are antiphase with respect to each other.

D: The air particles oscillate parallel to direction of propagation.

18.

Ans: C